

Tutorial 6

Vector spaces

1. *This question asks you to check the axioms for a vector space in a specific example. You will need to have your notes open at the definition of a vector space (page 91).*

Let $\mathbb{R}^2 = \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}\}$. We have seen that, with the usual definitions for addition and scalar multiplication of vectors, $\mathbb{R}^2$ is a vector space. In this question we use the usual scalar multiplication, but define a **different addition** to the usual one on $\mathbb{R}^2$ (we'll denote it by $\oplus$ to emphasize the difference). Your task will be to test whether $\mathbb{R}^2$ is a vector space with these two operations.

- Addition on $\mathbb{R}^2$ is defined as follows: For $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ in $\mathbb{R}^2$,

$$\mathbf{x} \oplus \mathbf{y} = (x_1 + y_1 + 1, x_2 + y_2 - 2).$$

- Scalar multiplication on $\mathbb{R}^2$ is defined as follows: For $\mathbf{x} = (x_1, x_2)$ in $\mathbb{R}^2$ and $\lambda \in \mathbb{R}$,

$$\lambda \mathbf{x} = (\lambda x_1, \lambda x_2).$$

- (a) Check that $\mathbf{x} \oplus \mathbf{y} \in \mathbb{R}^2$ and $\lambda \mathbf{x} \in \mathbb{R}^2$. (This is close to obvious; simply say why it is obvious.)

We now want to see whether $\mathbb{R}^2$ is a vector space over $\mathbb{R}$ when addition and scalar multiplication is defined as above. This will only be the case if *every one* of the eight axioms listed in Definition 2.1.4 for a vector space hold for $\mathbb{R}^2$ with the above addition and scalar multiplication.

- (b) Do the axioms 1 to 4 hold? If you claim an axiom holds, you must show that it holds (write out your check in full). If you claim that it does not hold, you must give an example to show that it does not hold.
[Hint: The zero vector and negatives may not be quite what you expect. Get your work for axioms 3 and 4 checked by a tutor.]
- (c) Do the axioms 5 to 8 all hold? Again, give proofs or counter-examples to support your claims.
- (d) Is $\mathbb{R}^2$ a vector space with the addition and scalar multiplication as defined above?

2. Let V be a vector space.
 - (a) Let $\mathbf{u} \in V$. Prove that $\mathbf{u} + \mathbf{u} = 2\mathbf{u}$. This is not one of the axioms of a vector space, so it *does* need proof. Be careful to use *only* the definition of a vector space in your proof; it is a good discipline to say at each step in your proof which axiom of the definition you are using. (*Hint: $\mathbf{u} = 1\mathbf{u}$.*)
 - (b) Prove Theorem 2.1.12(c), which says that in any vector space V , $\lambda\mathbf{0} = \mathbf{0}$ for any scalar λ . Again, be careful to use only the definition of a vector space in your proof. (*Hint: $\mathbf{0} = \mathbf{0} + \mathbf{0}$.*)
 - (c) Prove that in any vector space V , $\lambda\mathbf{u} = \mathbf{0}$ if and only if $\lambda = 0$ or $\mathbf{u} = \mathbf{0}$ (See Theorem 2.1.12(e)).
 - (d) Do the two statements you proved in (a) and (b) hold in the set $\mathbb{R}^2$ of Question 1, with the addition and scalar multiplication defined there? And the statement $\lambda\mathbf{u} = \mathbf{0} \Rightarrow \lambda = 0$ or $\mathbf{u} = \mathbf{0}$? Give a reason for your answer.
3. It follows from the definition of a vector space that every vector space V comes equipped with two operations, that of **addition** and **scalar multiplication**.
 - (a) Is it possible to use these two operations and the axioms of a vector space to define an operation of **subtraction** on V ? If so, how would you do it? (Your definition should work in any vector space, not just in a specific one.)
 - (b) Is the operation of subtraction you defined commutative?
 - (c) Is the operation of subtraction you defined associative?
 - (d) Prove that the additive identity of V is also a “subtractive identity”.